Networks and Security

Alireza Esfahani *BSc, MSc, PhD, FHEA, M IEEE, M ECSO*
Senior Lecturer in Cyber Security
Cyber Security Course Leader
School of Engineering and Computing
University of West London

UNIVERSITY OF WEST LONDON
The Career University

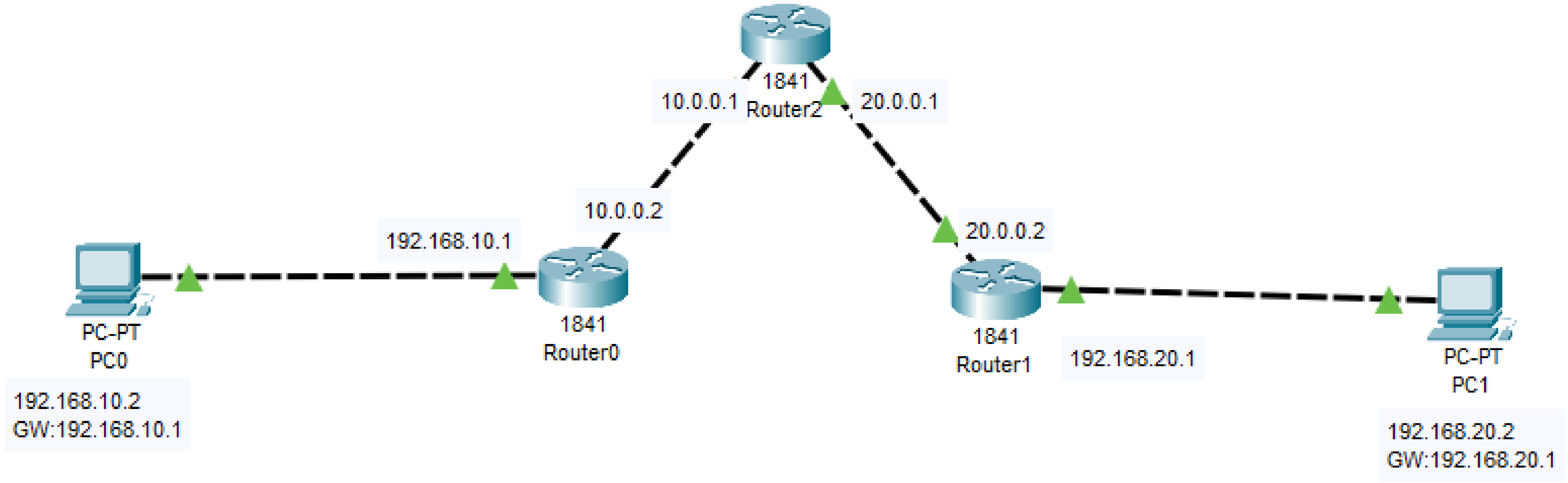
VPN-
Generic Routing Encapsulation (GRE) tunnel

Introduction

First, open the Cisco packet tracer desktop and select the devices given below:

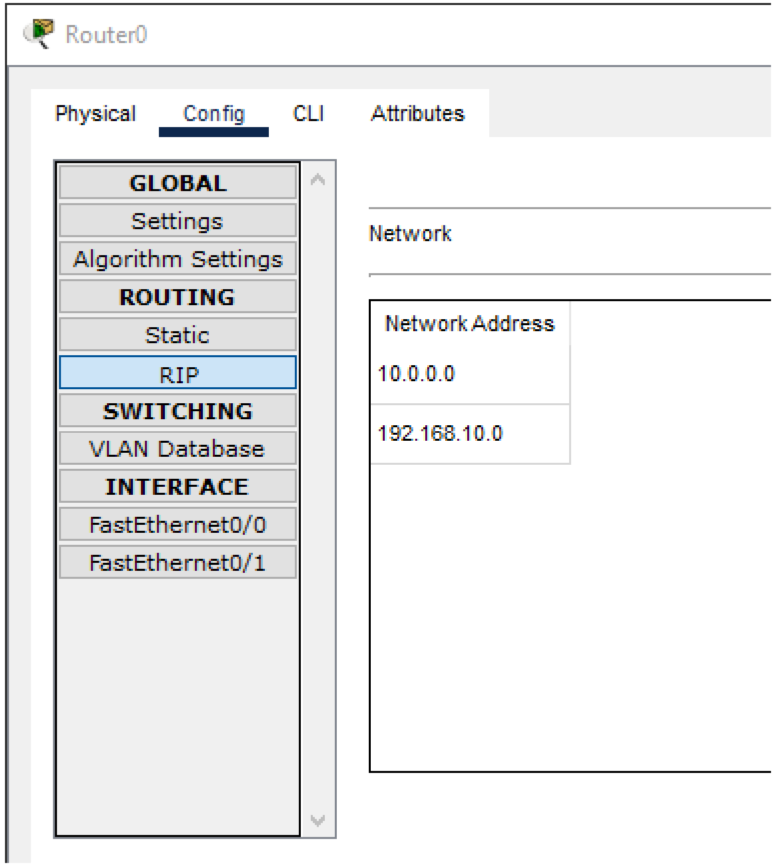
Device	Model Name	Quantity
PC	PC	2
Router	1841	3

Topology

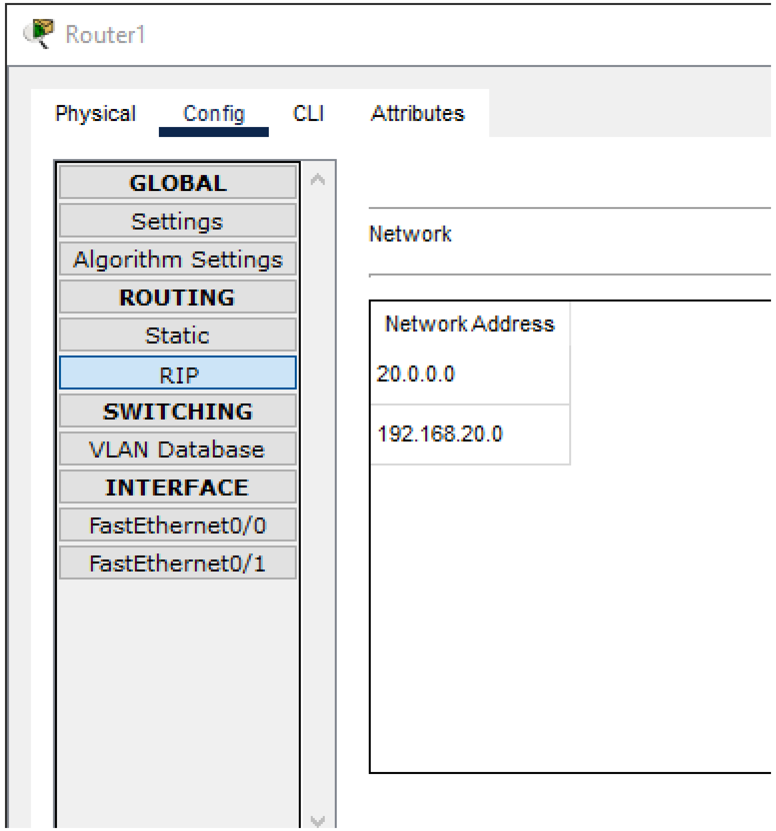


1841
Router2
10.0.0.1
20.0.0.1
10.0.0.2
20.0.0.2
192.168.10.1
1841
Router0
1841
Router1
192.168.20.1
PC-PT
PC0
192.168.10.2
GW:192.168.10.1
PC-PT
PC1
192.168.20.2
GW:192.168.20.1

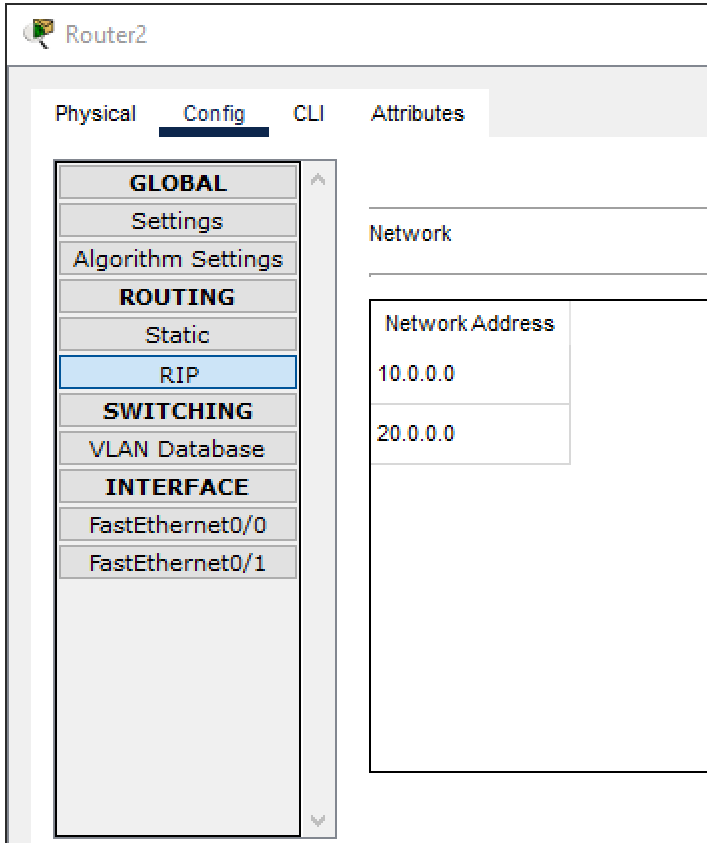
Apply RIP



Router 0



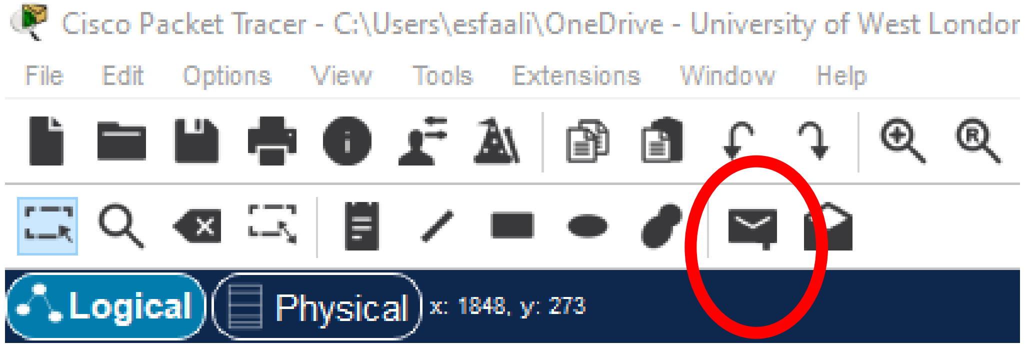
Router 1



Router 2

Check the status

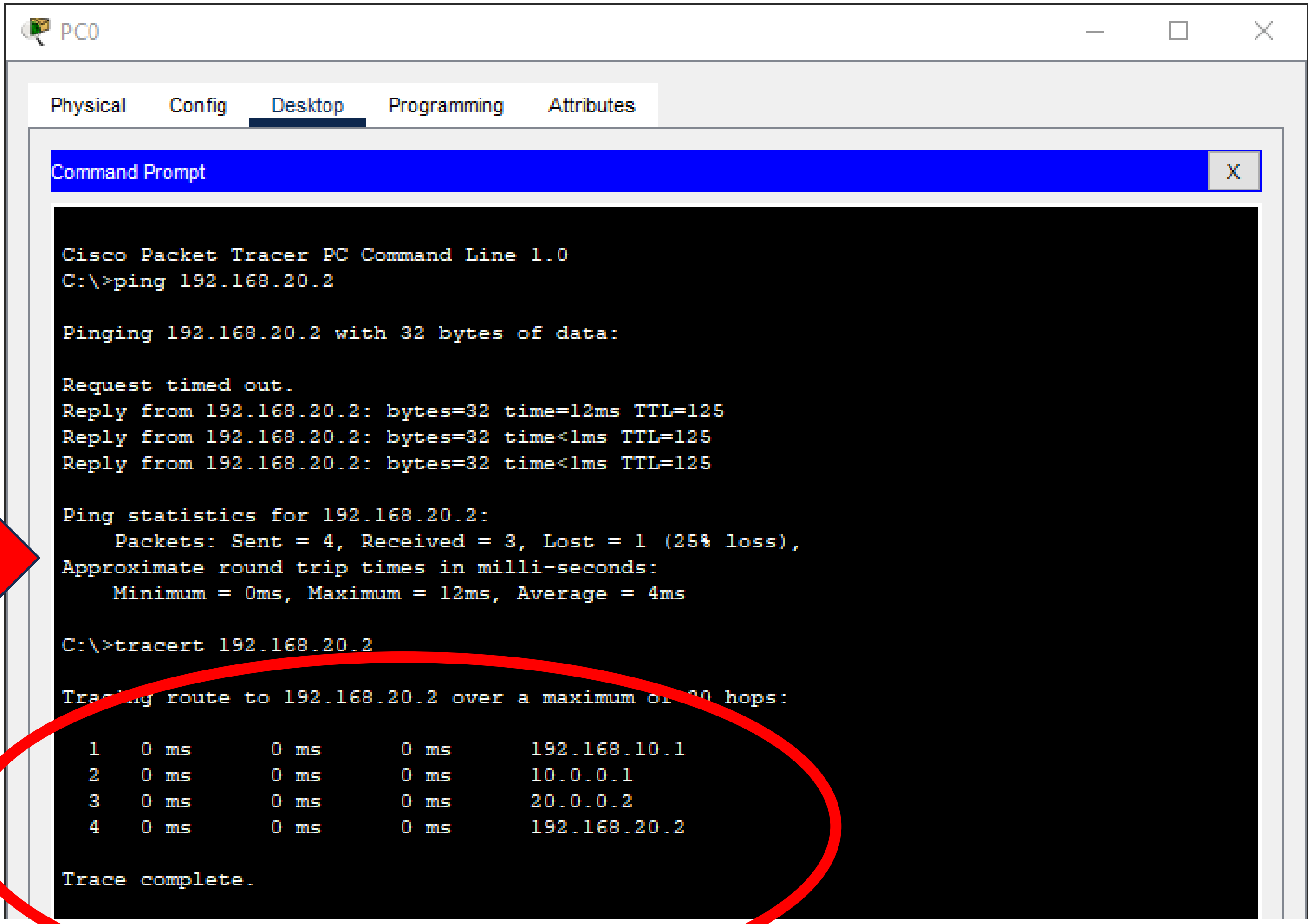
Click on and drop it at PC0 or PC1 to see if they are connected!



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	D
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	
	Successful	PC1	PC0	ICMP		0.000	N	1	(edit)	

Check the status

Click on PC0 and go to Desktop, and then use ping and tracert commands:

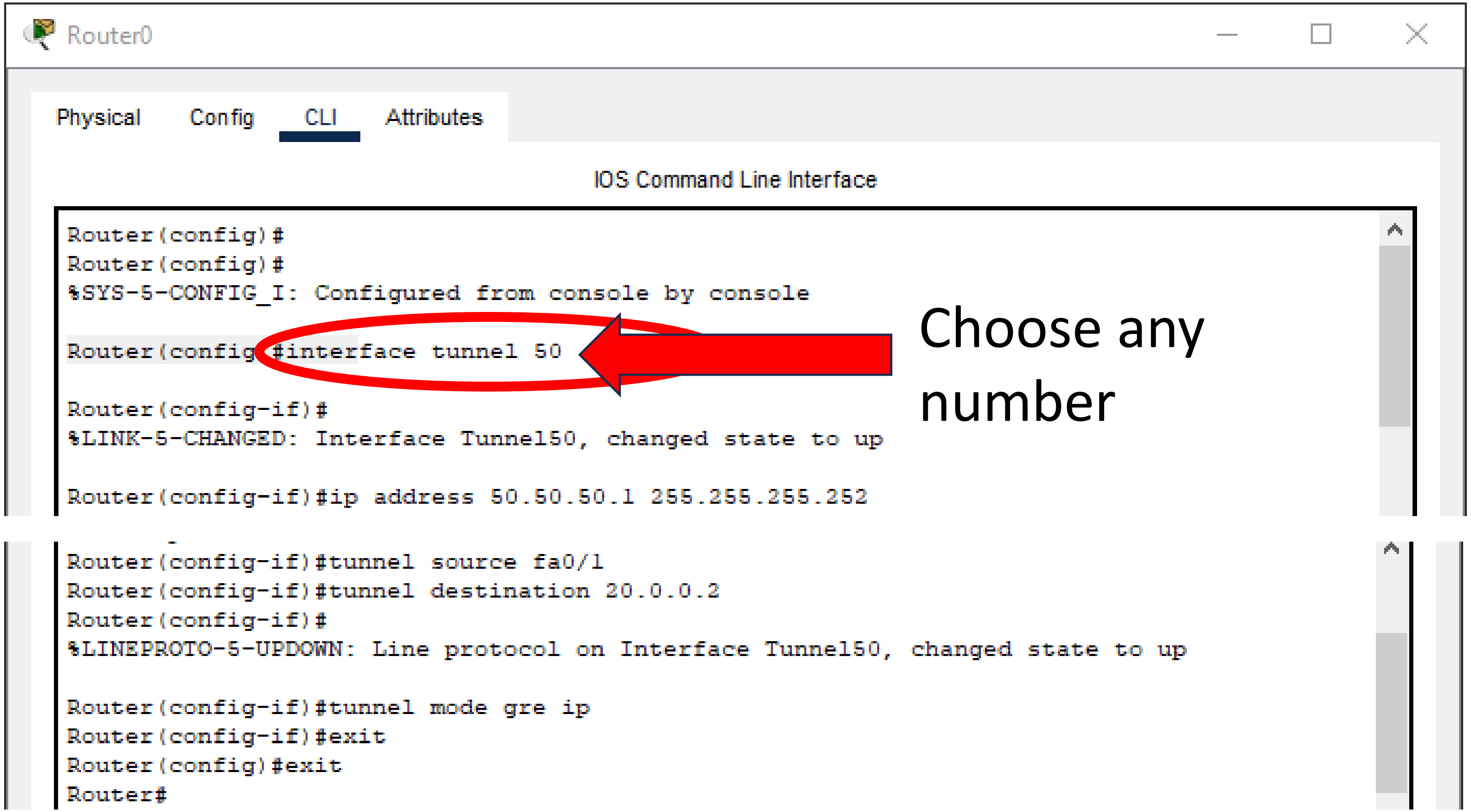


Why? Ask, if it’s not clear

Configure Tunnel for router 0

Tunnel set up

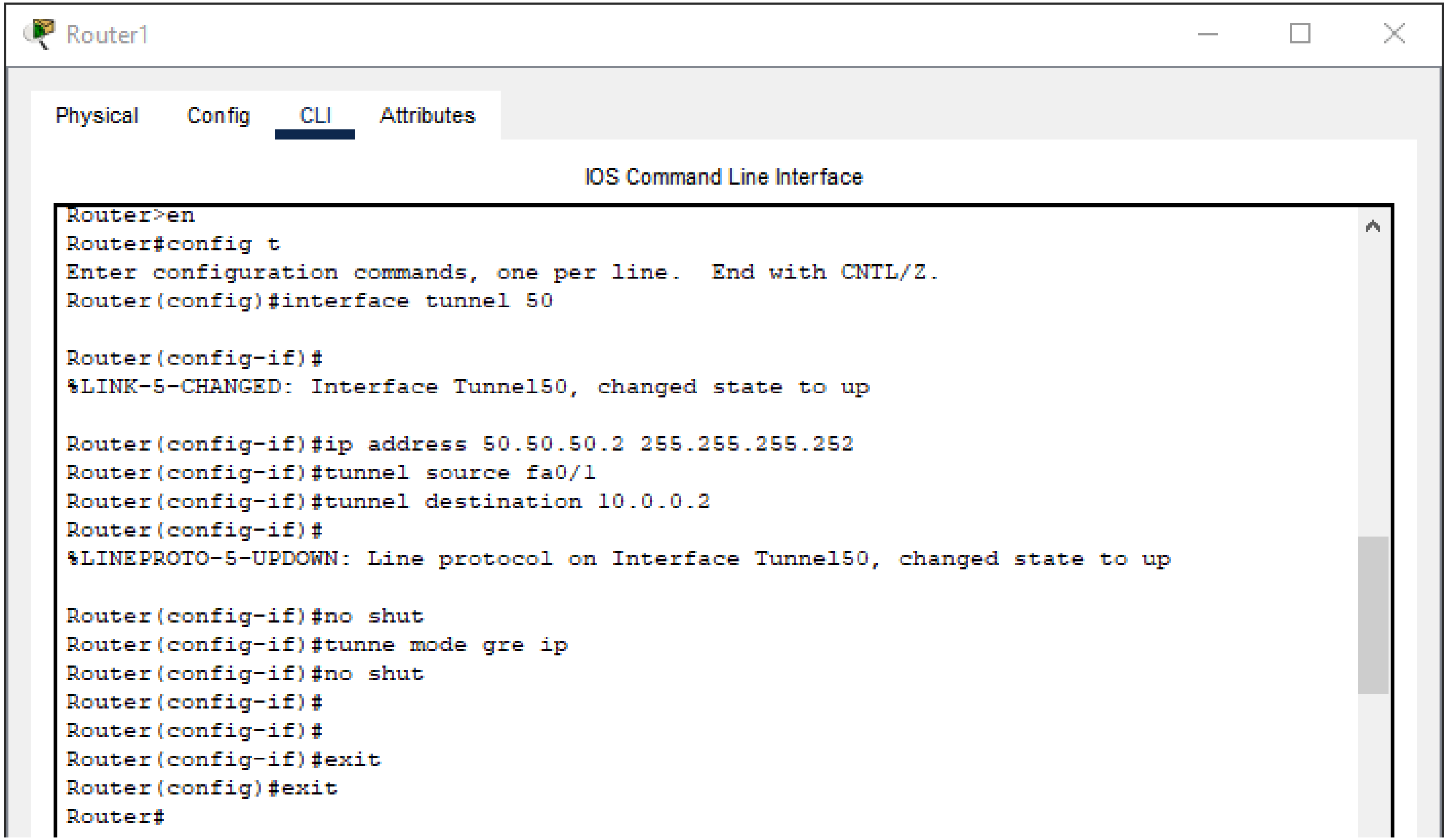
Follow the screenshot line by line



Configure Tunnel for router 1

Tunnel set up

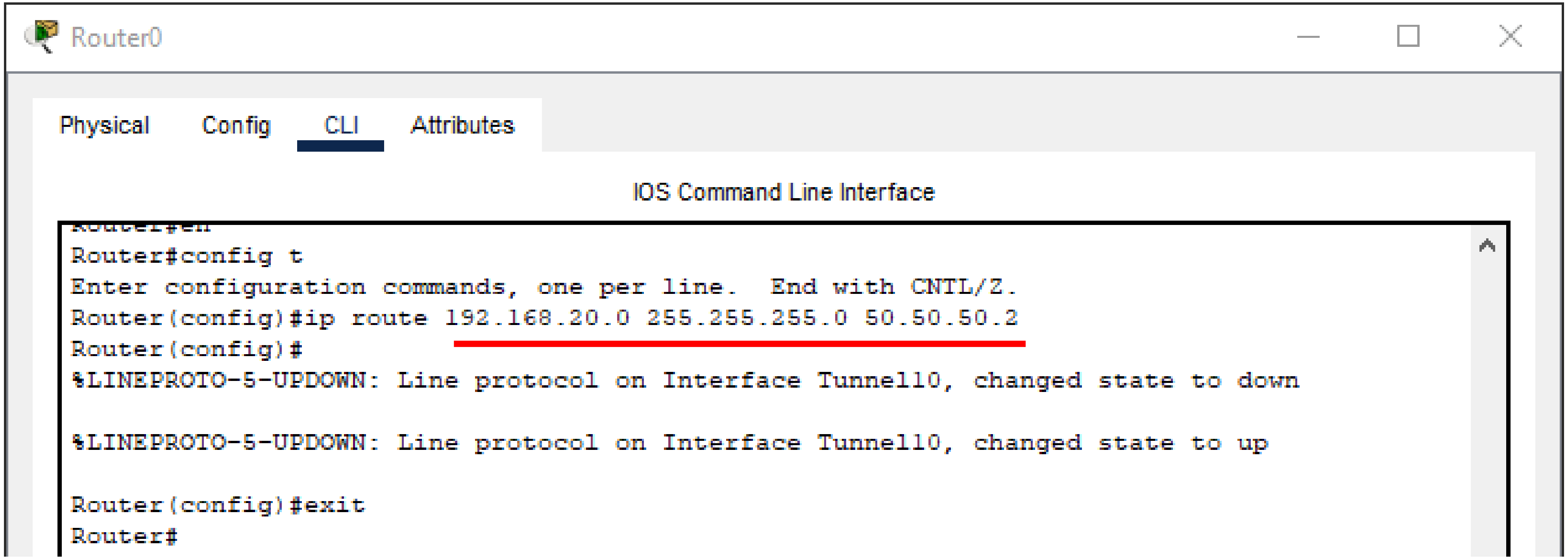
Follow the screenshot line by line



Configure routes for the Tunnel between routers

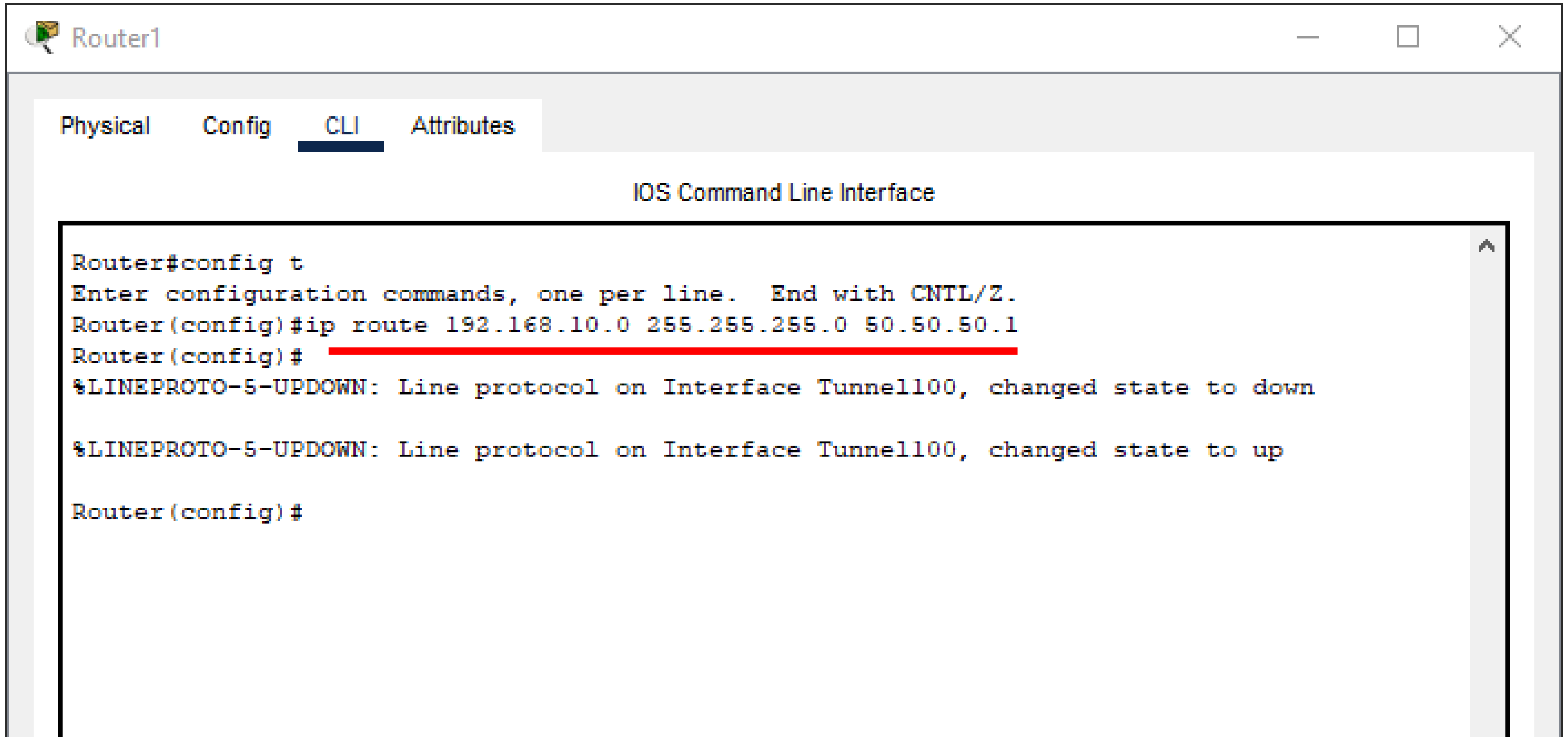
Router 0

Follow the screenshot line by line

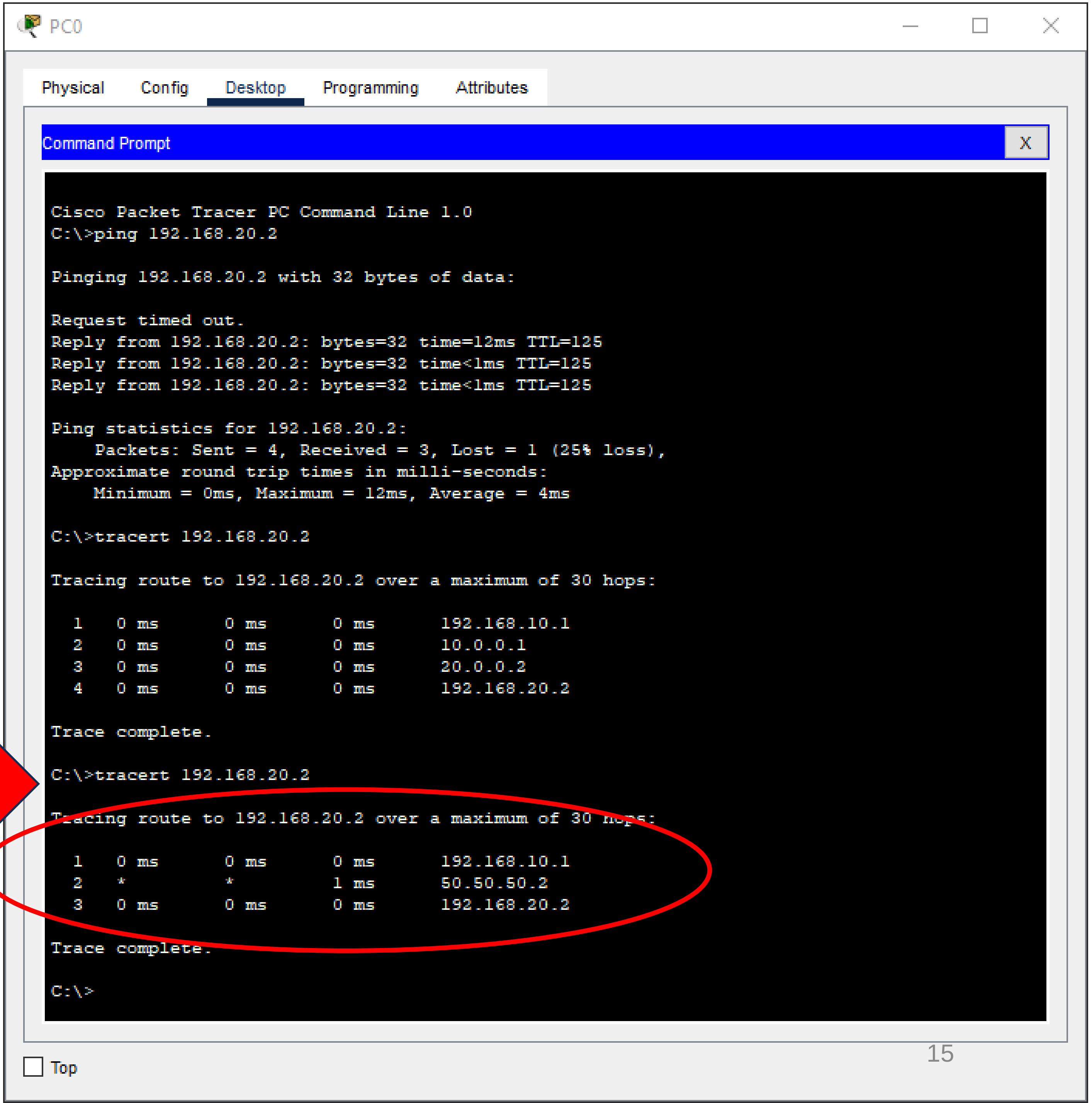


Router 1

Follow the screenshot line by line



Check the status



This is what you need to show us

Topology

This is what we created